

Caterina Paternoster

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EDUCATION

- Università Cattolica del Sacro Cuore** November 2021 – February 2025
International Ph.D. in Criminology Milan, Italy
Thesis Title: "Ties to Trust: Unveiling Patterns of Collaboration in 'Ndrangheta Clans".
Supervisors: Prof. Francesco Calderoni and Prof. Mario Andrea Lavezzi.
- University of Manchester** February 2024 – July 2024
Visiting Ph.D. student at the School of Social Sciences, Department of Criminology. Manchester, United Kingdom
- Università Cattolica del Sacro Cuore** October 2019 – September 2021
M.A. in Public Policies – Curriculum Security Policies Milan, Italy
Dissertation Title: "Beneficial Ownership Transparency and Anti-money Laundering: An Exploratory Analysis" - Supervised by Prof. Michele Riccardi. Final Grade: 110/110, *cum laude*.
- Università degli Studi di Padova** September 2016 – July 2019
B.A. in Political Science, International Relations, and Human Rights Padua, Italy
Thesis Title: "Missing Inflation: An Investigation into Its Determinant" - Supervised by Prof. Gianfranco Tusset. Final Grade: 110/110, *cum laude*
Awarded the *Mille e una lode* merit scholarship in 2019.
- Liceo Scientifico Leonardo da Vinci** September 2011 – July 2016
High School Diploma (EQF 4) Trento, Italy

WORKING EXPERIENCE

- Università Bocconi** January 2025 – Ongoing
Postdoctoral Research Fellow in Social Statistics Milan, Italy
 - Project NEMESIS - sociogeNEsis of criMinal nEtworks: reconStruction, dIScovery and diSruption. PI: Daniele Durante
- Transcrime - Joint Research Centre on Innovation and Crime** November 2021 – December 2024
Researcher Milan, Italy
- Crime&Tech Srl** February 2021 – September 2021
Internship Milan, Italy
- Comune di Trento** September 2018 – October 2018
Internship Trento, Italy

PROJECTS

Selected projects conducted as Researcher at Transcrime (2021–2024)

- FALCON** 2023 – December 2024
Fight Against Large-scale Corruption and Organized Crime | Horizon Europe
Contribution: Coordinated project activities on behalf of Transcrime; co-authored policy reports; delivered training sessions; presented findings at international and consortium conferences.
- KLEPTOTRACE** January 2024 – July 2024
Strengthening EU asset recovery and sanction tracing against transnational high-level corruption | ISF
Contribution: Investigated transnational high-level corruption and sanction evasion through case collection; contributed to policy reports.

• ALLIES

2023 – 2024

AI-based Framework for Supporting Micro and Small HSPs on the Report and Removal of Online Terrorist Content
| ISF

Contribution: Conducted qualitative analysis through literature review and case analysis on online terrorist radicalization; supported in developing taxonomy and risk assessment module; co-authored policy reports and scholarly papers; presented findings at international and consortium conferences.

• CEASEFIRE

2022 – January 2024

Advanced Versatile Artificial Intelligence Technologies and Interconnected Cross-sectoral Fully Operational National Focal Points for Combating Illicit Firearms Trafficking | Horizon Europe

Contribution: Conducted and developed the criminological analysis on the trends and *modi operandi* of firearms trafficking; co-authored policy reports and scholarly papers; presented findings at international and consortium conferences.

• CSABOT

2021 – 2023

Civil Society Advancing Beneficial Ownership Transparency, Preparatory action - Capacity Building, Programmatic Development and Communication in the Context of the Fight Against Money Laundering and Financial Crimes | Horizon Europe

Contribution: Conducted qualitative and quantitative analysis on the risk associated with the lack of ownership transparency in the AML domain, summarized in a co-authored handbook; contributed to policy reports and scholarly papers; delivered trainings to civil society from selected EU member states; presented findings at consortium conferences.

• ANAC

2022

Identification and Evaluation of Conflicts of Interest | PON 'Governance e Capacità Istituzionale 2014-2020'

Contribution: Developed the methodology to identify economic interest and relevant links of individuals subject to declaration of absence of conflicts of interest; contributed to the exploratory quantitative analysis in a sub-sample of *responsabili per la prevenzione della corruzione e della trasparenza*, operating in local administrations; co-authored the final report.

• EBOCS III

2021

European Business Ownership and Control Structures | ISF

Contribution: Analyzed data stemming from official beneficial ownership registers across some selected EU members states; co-authored the final report.

TEACHING

• Corruption, Compliance and Risk Control

February 2026-ongoing

Adjunct Professor - 56h

Master Degree in Security, Deviance, and Risk Control, Università Milano-Bicocca, Milan

• Data Science Laboratory

6-17 July 2026

Instructor - 12h

Bocconi Summer School for High School Students, Università Bocconi, Milan

• Quantitative Methods for Social Sciences (Data Analytics)

February 2026-ongoing

Teaching Assistant - 25h

Bachelor Degree in Data, Society & Organisations HEC-BOCCONI, Università Bocconi, Milan

• Methods and Techniques of Criminological Research

2024

Teaching Assistant - 5h

Master Degree in Public Policy - Crime and Security Analysis, Università Cattolica del Sacro Cuore, Milan

OTHER QUALIFICATIONS AND EXPERIENCES

• Cultore della Materia - Università Cattolica del Sacro Cuore

2022-2024

- Models Applied to Crime Analysis | Faculty of Political and Social Science
- Methods and Techniques of Criminological Research | Faculty of Political and Social Science

• Peer Tutor | Funding *Mille e una lode* - Università degli Studi di Padova

2019

- Tutoring of two international students from Haiti in their studies and exam preparation (50h)

PUBLICATIONS

UNDER REVIEW

- Diviak T, **Paternoster C**, Jürgen L, Calderoni F. (2026+) Exploring the Dynamic Interplay between Communication and Co-offending Using Relational Hyperevent Data on an Italian Mafia Network. Under peer review at *Journal of Quantitative Criminology*.
- Screen P, **Paternoster C**. (2026+) Methodological Approaches and Challenges in Mapping Cross-Border Drug Flows and Their Determinants. Under peer review at *Oxford Intersections*.
- **Paternoster C**, Calderoni F. (2025+) Ties to Trust: Factors Influencing Collaboration Within and Across Organised Criminal Groups. Conditionally accepted at the *British Journal of Criminology*.

JOURNALS

- Carbone C, **Paternoster C**, Riccardi M. (2025) Ville Lumière Ou Ville Obscure? Assessing the Secrecy of Firms Owning Real Estate Properties in Paris. *Trends in Organized Crime* <https://doi.org/10.1007/s12117-025-09570-8>
- **Paternoster C**, Nazzari M, Jofre M, Uberti T.E. (2025) Inside the Leak: Exploring the Structure of the Conti Ransomware Group. *Global Crime* 26(2), 148–171. <https://doi.org/10.1080/17440572.2025.2473350>
- Mademlis I, Mancuso M, **Paternoster C**, Evangelatos S, Finlay E, Hughes J, Radoglou-Grammatikis P, Sarigiannidis P, Stavropoulos G, Votis K, Papadopoulos G Th. (2024) The invisible arms race: digital trends in illicit goods trafficking and AI-enabled responses. *IEEE Transactions on Technology and Society* 6(2), 181-199. <https://doi.org/10.1109/TTS.2024.3514683>

BOOK CHAPTERS

- Kalpakis G, **Paternoster C**, Mancuso M, Kozhuharova D, Spathi T, Semertzidis T, Tsikrika T, Vrochidis S. (2024) AI-Based Framework for Supporting Micro and Small Hosting Service Providers on the Report and Removal of Online Terrorist Content. In: Gkotsis, I., Kavallieros, D., Stoianov, N., Vrochidis, S., Diagourtas, D., Akhgar, B. (eds), *Paradigms on Technology Development for Security Practitioners. Security Informatics and Law Enforcement*. Springer, Cham. https://doi.org/10.1007/978-3-031-62083-6_20
- Dugato M, **Paternoster C**. (2022) Organized Crime and the Retail Sector. In Zabyelina, Y., Thachuk, K. L. (eds), *The Private Sector and Organized Crime: Criminal Entrepreneurship, Illicit Profits, and Private Sector Security Governance*, Routledge: 180- 194. 10.4324/9781003198635-13

CONFERENCE PAPERS

- Bianchi S, Mancuso M, **Paternoster C**, Kalpakis G, Tsikrika T, Vrochidis S, Kozhuharova D, Jaeger B. (2023) Artificial Intelligence to Counter Cyber-Terrorism. *Proceedings of the International Conference on Cybersecurity and Cybercrime (IC3)*, 10, 12-20 <https://doi.org/10.19107/CYBERCON.2023.02>

POLICY REPORTS AND PROJECT DELIVERABLE (SELECTION)

- Nicolazzo G, Bernatskyi B, Ventre L, Cascone G, **Paternoster C**, Ovsianikova I, Bremi D, Axinte L. (2025) Restrictive measures: the challenges in front of the EU. Report on schemes employed to circumvent sanctions. Milan: Transcrime - Joint Research Centre on Innovation and Crime. ISBN: 978-88-99719-50-0
- Cascone G, **Paternoster C**, Riccardi M, Poltoratskaya V, Tóth B, Baez-Camargo C, Costa J, (2025). High-level Corruption: an Analysis of Schemes, Costs and of Policy Recommendations. Policy Brief of Project FALCON. Milan: Transcrime – Joint Research Centre on Innovation and Crime. ISBN: 978-88-99719-48-7
- **Paternoster C**, Riccardi, M, Cascone, G, Ventre, L, Anastasio, M, Nicolazzo, G, Venturini, F, Poltoratskaya, V, Toth, B, & Costa, J. (2024). Facilitators, schemes, interlinks and modi operandi of corruption. D2.1 FALCON project - Fight Against Large-scale Corruption and Organized Crime. *Available upon request*
- Nicolazzo G, **Paternoster C**, Cascone G, Ventre L, Anastasio M, Bernatskyi B, Riccardi M, Rondani C. (2024) Handbook on new forms of, and risk factors for, high-level transnational corruption schemes. Milan: Transcrime - Joint Research Centre on Innovation and Crime. ISBN: 9788899719494
- **Paternoster C**, Mancuso M. (2023). Report on online radicalisation, extremism and terrorism. ALLIES project - AI-based framework for supporting micro and small HSPs on the report and removal of online terrorist content. *Available upon request*
- Carbone C, **Paternoster C**, Riccardi M (2023). Beneficial owners of European companies (and related risks). In European Commission. Directorate General for Financial Stability, Financial Services and Capital Market Union. (Eds), *Preparatory action, Capacity building, programmatic development, and communication in the context of the fight against money laundering and financial crimes. Final report annexes (p 3-62)* Publications Office.
- Dugato M, **Paternoster C**, Favarin S (2021). La Sicurezza nel Retail in Italia 2021. Milan: Crimotech (Spin-off di Università Cattolica del Sacro Cuore-Transcrime). ISBN: 978-88-99719-27-2

CONFERENCES AND WORKSHOPS

- **Illicit Networks Workshop** June 29-30, 2026
Presentation: Paternoster C. "Structural Patterns in Interfamily Marriages in the 'Ndrangheta". Vienna, Austria
- **Tenth International Workshop on Social Network Analysis (ARS'25)** October 30-31, 2025
Presentation: Paternoster C, Calderoni F. "Ties to Trust: Unveiling Patterns of Collaboration in 'Ndrangheta Clans". Naples, Italy
- **European Society of Criminology (ESC) Annual Meeting** September 03-06, 2025
Presentation: Paternoster C, Calderoni F. "Ties to Trust: Unveiling Patterns of Collaboration in 'Ndrangheta Clans". Athens, Greece
- **American Society of Criminology (ASC) Annual Meeting** November 13-16, 2024
Presentation: Paternoster C. "Cohesion and Connection: An Analysis of 'Ndrangheta Clans Interactions." San Francisco, CA, USA
- **European Society of Criminology (ESC) Annual Conference** September 11-14, 2024
Presentation: Paternoster C, Anastasio M, Nicolazzo G, Ventre L, Cascone G, Venturini F, Riccardi M. "Modus Operandi of Corruption: Examining Drivers, Actors, and Financial Flows in Three Typologies." Bucharest, Romania
- **American Society of Criminology (ASC) Annual Conference** September 15-18, 2023
Presentation: Paternoster C, Nazzari M, Jofre M, Uberti TE. "Network Analysis for Cybercrime: Breaking Down the Structure of the Conti Ransomware Group." | | Paternoster C, Calderoni F. "Collaboration in Organized Crime: Unveiling the Role of Family Ties". Philadelphia, PA, USA
- **European Society of Criminology (ESC) Annual Conference** September 6-9, 2023
Presentation: Paternoster C, Mancuso M. "Patterns of Online Radicalization: A Study of Jihadist and Far-Right Extremist Groups". Florence, Italy
- **New Challenges for the Criminal Investigation of Organized Crime** May 11-12, 2023
Presentation: Paternoster C, Nazzari M, Jofre M, Uberti TE. "Network Analysis for Cybercrime: Breaking Down the Structure of the Conti Ransomware Group." Porto, Portugal
- **Workshop - Training delivered to Fincantieri S.p.A.** June 28, 2023
Training delivered by Carbone Carlotta and Paternoster Caterina on the "Analysis of Ownership Structures and Identification of Beneficial Ownership." Online
- **Curbing Financial Crimes in Europe with Beneficial Ownership Transparency** November 29-30, 2022
Final Conference of the CSABOT project. Brussels, Belgium

COMPETENCES

- **Office Tools:** Proficient with Microsoft Office (Word, Excel, PowerPoint).
- **Data Analysis:** R Studio
- **Network Analysis:** R packages, UCINET.
- **Spatial Analysis:** R packages, Qgis.

LANGUAGES

- **Italian:** Native proficiency.
- **English:** Professional working proficiency (C1) - Advanced written and verbal communication skills.
- **Spanish:** Basic user.